

Bi-ETC: A Bidirectional Encrypted Traffic Classification Model Based on BERT and BiLSTM

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Abstract—While packet encryption technology provides user privacy protection, it also brings challenges to traffic classification. Traditional traffic classification technologies based on packet fields or payload data content are becoming powerless in the face of encrypted traffic. Although traffic encryption technology can hide data content, it cannot change the inherent characteristics of application traffic. Therefore, how to classify encrypted traffic by mining traffic features has become a topic of widespread concern. An enormous amount of research shows that machine learning algorithms achieve good results in encrypted traffic classification. However, there is still room for improvement. For example, deep learning methods require a large amount of labeled training data, and it is difficult for pre-trained models to capture the associated features of long sequences of packet. This paper propose a Bidirectional encrypted traffic classification model based on BERT and BiLSTM(Bi-ETC), by inserting [CLS]Token at the connection of Token sequence between BERT and BiLSTM, strengthens BiLSTM's focus on packet-level features while maintaining BiLSTM's capture of Token sequence context. Experimental results show that the Bi-ETC model is superior to all proposed classification methods on the ISCX VPN dataset, reaching 99.43% in F1 score and 99.7% in accuracy.

Index Terms—Privacy Protection, Network communication security, Encrypted traffic classification, BERT, BiLSTM

I. INTRODUCTION

In recent years, with the widespread need to protect data transmission and user privacy, protocols and applications are more inclined to adopt encryption methods. Therefore, how to realize the accurate identification of network encrypted traffic is an important issue in the field of cyberspace security, and is also the focus of current network behavior analysis, network planning and construction, network anomaly detection and network traffic model research[1]. Encrypted traffic is growing in a large number of current communication networks, and a variety of encryption protocols have been used[2], such as IPSec, SSL/TLS, SSH, BitTorrent, Skype, etc. Different encrypted protocol packets are also located in different network

protocol layers, which makes the classification of encrypted traffic difficult. Even if the same encryption protocol is used, the data distribution of encrypted traffic may show completely different characteristics due to the different original traffic distribution[3]. Therefore, the traffic classification method for a certain type of encrypted traffic cannot adapt well to new environments or unseen encryption policies.

The traditional port-based traffic identification method[4] is no longer reliable in the classification of encrypted traffic, and can only be used as an auxiliary means for traffic classification. In addition, earlier works utilize the remaining plaintext in encrypted traffic to build fingerprints and perform fingerprint matching classification[5] are not applicable to emerging encryption technologies, as plaintext becomes more sparse or obfuscated. To this end, many scholars use machine learning[6, 7] to assist traffic classification methods, and train classifiers based on manually extracted packet or its statistical features at the data flow level. Since these features require prior knowledge and experience, they are also time-consuming to extract. More importantly, there is no guarantee that these features are really helpful to improve the classification performance. As a result, many researchers have utilized deep learning methods[8, 9] to automatically learn complex patterns from raw traffic and achieved significant performance gains. However, these methods require a large amount of labeled training data to achieve good performance.

In recent years, pre-trained models have made significant breakthroughs in natural language processing[10], computer vision[11] and other fields. The advantage of pre-trained models is that they can learn the representation of the data itself from a large amount of unlabeled data and only need to fine-tune on a small amount of labeled data. So that the downstream tasks can better learn their own features and the required knowledge. At the same time, pre-training methods are also widely used in network communication scenarios, and [12, 13, 14] proposed the application of pre-training methods in solving encrypted traffic classification problems. However,

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current research cannot capture the long-distance contextual relationships in encrypted traffic Token sequences. In this paper, we propose to use Bi-ETC for encrypted traffic classification tasks, which can not only capture the long-distance contextual relations in traffic sequences, but also enhance the model's attention to traffic semantic features. The specific contributions of this paper can be summarized as follows:

- According to the state of art we know we first propose to use the BiLSTM(Bidirectional Long Short Term) model in encrypted traffic classification to capture the long-distance dependencies well between the BERT(Bidirectional Encoder Representations from Transformers) output Token sequence features.
- we propose to insert packet-level [CLS]Token semantic features at the forward and backward starting positions of BiLSTM, so as to enhance BiLSTM's attention to packet-level features.
- Experiments show that our method achieves excellent results on publicly available encrypted traffic classification datasets, and outperforms eleven state-of-the-art models in terms of precision, recall, and F1-score.

The rest of this paper is organized as follows. In Section II, we review some of the most important and recent research on traffic classification. In Section III, we introduce the motivation and objective of the model. Section IV introduces our proposed method, namely Bi-ETC. In Section V, the experimental results and application identification performance are analyzed. Finally, we conclude the paper in Section VI.

II. RELATED WORK

In recent years, network traffic classification has become a research hotspot in academia and industry[15]. With the rapid growth of encrypted traffic in the network, traditional traffic classification methods based on ports[16] and deep packet inspection[17] can no longer meet the current requirements. Machine learning has been widely used for identification and classification of encrypted traffic, and FlowPrint[18] is a semi-supervised method for fingerprinting mobile applications from encrypted network traffic. AppScanner[19] utilizes the statistical characteristics of packet size to train a random forest classifier, while BIND[20] also utilizes statistical characteristics of temporality.

Recently, Aceto et al.[21] proposed a method combining the advantages of deep learning and big data to solve the problem of network traffic classification. Rezaei et al.[22] outlined a general framework for deep learning-based traffic classification, and introduced commonly used deep learning methods and their applications in traffic classification tasks. Traffic classification based on deep learning can be traced back to 2015[23], where Wang et al. used a simple SAE autoencoder. Lotfollahi et al.[8], after an initial preprocessing stage of the data, feed packets into a deep packet framework that embeds stacked autoencoders and convolutional neural networks to classify network traffic. Zeng et al.[24] proposed a novel DL-based framework capable of classifying encrypted traffic and detecting malware traffic. Yao et al.[25] modeled

time-series network traffic through a RNN(recurrent neural network) and introduced an attention mechanism. Ren et al.[26] proposed a tree-structured recurrent neural network (tree-RNN), setting a specific classifier for each small classification. Liu et al.[9] proposed Flow Sequence Network(FS-Net), which can deeply mine the latent sequence features of flows, and introduced a reconstruction mechanism that can improve feature effectiveness.

PERT[12] uses the state-of-art dynamic word embedding technique to perform automatic traffic feature extraction and provides a traffic classification framework in which unlabeled traffic is used to pre-train an encoding network that learns the contextual distribution of traffic payload bytes. Then, the pre-trained network is reused for downward classification to obtain the enhanced classification results. CBD[26] has the generalization ability to classify encrypted traffic in the real environment, and the overall structure of CBD can be described as the encoder of a one-dimensional CNN and a Transformer. The model classifies encrypted traffic from the packet level and the traffic level. ET-BERT[13] pre-trains deep contextualized datagram level representations from large-scale unlabeled data, and the pre-trained model can fine-tune on a small amount of task-specific labeled data. ET-BERT can effectively learn implicit relationships in unlabeled traffic, thereby improving the effect of traffic classification in different scenarios. Compared with ET-BERT[13], BFCN[14] captures byte-level local features in traffic through convolution operation.

III. MOTIVATION AND OBJECTIVE

Recently, pre-training has also been introduced into encrypted traffic classification[12, 13, 14] and achieved excellent performance on multiple datasets. However, [12] lacks pre-training tasks designed for traffic and reasonable input to show the effect of pre-training models. [13] can better represent encrypted traffic, but cannot capture the relationship between encrypted traffic Tokens. [14] Reasonable use of CNN to capture the relationship between encrypted traffic Tokens, but cannot capture long-distance contextual relationships from encrypted traffic Token sequences. In encrypted traffic classification, although the payload has no semantics, Sengupta et al.[27] proposed that the randomness difference between different ciphertexts can be used to distinguish different applications, which shows that encrypted traffic is not completely random, and there is an implicit pattern. Therefore, we need to use the BERT module to extract the implicit features of encrypted traffic and obtain the feature vector of each encrypted traffic Token. And because the extracted feature vector of each encrypted traffic Token has the same timing relationship as the original encrypted traffic sequence, in order to solve the problem of timing relationship in the encrypted traffic sequence, this paper introduces the BiLSTM model based on the BERT pre-training model. Due to the long byte sequence of the data packet, the BiLSTM model alleviates the problem of gradient disappearance and gradient explosion compared with the traditional RNN. The ability to model the global context

of data. Therefore, by using BiLSTM, we can better process sequence data and capture contextual semantic features and timing features between Token feature representations.

IV. METHODOLOGY

A. Data Pre-processing

In order to improve data quality and improve training efficiency, we will first filter out retransmission and out-of-sequence data packets and network query data packets (eg, ICMP, DNS, etc.), and only keep IP data packets. Because re-transmissions and out-of-sequence packets are usually caused by abnormal network transmissions, they may distort traffic patterns and affect classification accuracy[8]. By removing these packets, you can ensure more accurate and consistent traffic patterns in your training and test datasets. However, network query data packets are usually caused by network requests such as domain name resolution, advertisement tracking, and statistical analysis, and they have nothing to do with the purpose of traffic classification. Retaining only IP packets can reduce the size of the data set and reduce unnecessary processing and analysis during training and testing, thereby improving efficiency.

Since our approach does not rely on any visible plaintext information, we will only intercept the transport layer payload of each packet and discard the rest (e.g., Ethernet headers, IP headers, transport layer headers, etc.). The payload is then converted to hexadecimal encoding for processing to ensure data consistency and commonality, and to avoid character encoding issues. Then, the hexadecimal payload is encoded by the bi-gram model, which is converted into a Token sequence to help capture the context between adjacent bytes.

1) labeling dataset

as mentioned in the previous section, the pcap files in the dataset are labeled based on the application and the network activity of the application, so in order to identify the application type, we need to redefine the label. We redefined the pcap files collected during VPN sessions into 17 label categories, as shown in Table 1 below. Then, a maximum of 5000 samples were selected from each type of data, among which the number of samples in AIM_Chat and ICQ categories were 1340 and 823, respectively, to test the effect of our method on the label imbalanced data set. Finally, we divided the dataset into training, validation and testing datasets according to the ratio of 8:1:1.

B. Model architecture

In this section, we will explain our proposed Bi-ETC, whose network structure is shown in Figure 1 below. The model mainly consists of a feature extraction module based on BERT and one based on BiLSTM. The model is composed of the up-down sequence text encoding module of Memory. Specifically, we first input the pre-processed encrypted traffic tokens into BERT, and use the BERT feature extraction module to extract the token-level features of encrypted traffic. And before each encrypted traffic Token sequence, the [CLS]Token

TABLE I
DATASET LABEL

label	class	count
1	AIM_Chat	1340
2	Email	5000
3	Facebook	5000
4	Gmail	5000
5	Hangout	5000
6	ICQ	823
7	Netflix	5000
8	SCP	5000
9	Skype	5000
10	Spotify	5000
11	Tor	5000
12	Torrent	5000
13	Vimeo	5000
14	VoipBuster	5000
15	VPN-FTPS	5000
16	VPN-SFTP	5000
17	Youtube	5000

label is introduced. By learning to model the semantic correlation between the representation of this special Token label and the entire encrypted traffic Token sequence, the BERT model obtains a feature vector representing the entire encrypted traffic packet. Therefore, the features extracted by the [CLS]Token label can be considered as the features that represent the whole packet level and are the most classification meaningful features. Similarly, other ordinary encrypted traffic Token sequences have also been extracted with corresponding features by BERT. In addition, although BiLSTM performs better than LSTM(Long Short Term Memory networks) and traditional RNN models on long sequences, there is still the phenomenon of forgetting [28]. Therefore, the [CLS]Token sequence of the head is copied to the end of the BiLSTM input sequence, and for both forward and backward LSTM, the [CLS]Token feature can be recalled again at the end of the sequence to alleviate the forgetting of the most important packetlevel feature information. Moreover, this can better capture the effective classification features in the [CLS]Token to achieve the purpose of accurate classification. In addition, we adopt the dropout technique for the output of BiLSTM, which reduces the complexity and capacity of the neural network by randomly setting the output of some neurons to zero during the training process, thereby reducing the risk of overfitting. Finally, we concatenate the output features of the last layer of the forward and backward LSTM as the final classification features and put them into the SoftMax network for classification to obtain the final classification target.

It is worth mentioning that in the experimental part, we did two ablation experiments to verify the performance of the model. First, compare the accuracy and loss of the two strategies whether to insert [CLS]Token at the end of the

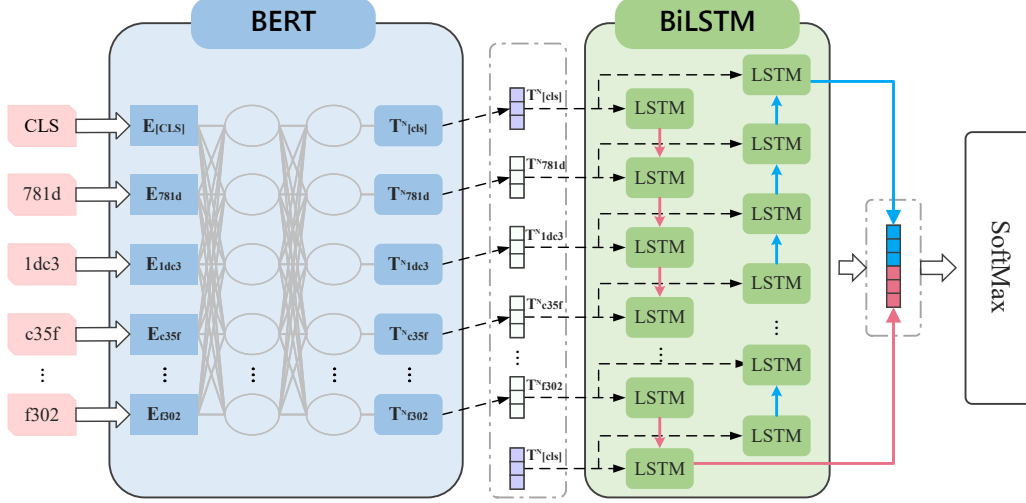


Fig. 1. Architecture of Bi-ETC model.

BiLSTM input sequence to verify the advantages of the model. Second, the effects of two different activation functions, tanh and PReLU, on the output features of BiLSTM were compared in SoftMax to verify the advantages of PReLU in this classification task.

Corresponding to $\mathbf{T}^N_{[\text{cls}]}$, $\mathbf{T}^N_{[781d]}$, $\mathbf{T}^N_{[1dc3]}$, $\dots$, including [CLS]) trained by BERT's hidden layer by a learnable weight $W_a \in R^{d_a \times n}$ as the input of the BiLSTM pattern. The formula is as follows:

$$a_i = g_1(W_a C_i + b_a) \quad (1)$$

where $1 \leq i \leq n$, n denotes the dimension of the feature vector output after fine-tuning of the BERT model, $a_i \in R^{d_a}$ is the input vector of the BiLSTM layer, b_a is the offset vector of dimension d_a . Here, we adopt Sigmoid as the activation function g_1 .

The ordinary LSTM model calculates the one-way hidden layer sequence h , while BiLSTM calculates the forward hidden layer vector as $\vec{h}_i$ and the backward hidden layer vector as $\overleftarrow{h}_i$ and finally combines the two for output vector vi . The calculation formula is as follows:

$$v_i = \vec{h}_i + \overleftarrow{h}_i \quad (2)$$

In addition, the hidden layer of the model as follows:

$$h_i^d = g_2(W_h^d a_i + U h_{i-1}^d + b_h^d) \quad (3)$$

Where $W_h^d a_i \in R^{d_h \times d_a}$ is the weight matrix of a_i , $d \in \{0, 1\}$ represents two different directions in the hidden layer, and U is the weight matrix of the output sequence h^d of the hidden layer at time i . h_{i-1}^d corresponds to the hidden layer output sequence at the previous time $i-1$; b_h represents the offset vector in the d direction. Then, the output sequences b_h^d of all

hidden layers are combined to obtain the feature vector H at the end of the set. Then, the feature vector H is transferred to a fully connected layer with PReLU function to classify the encrypted traffic.

V. EXPERIMENTS AND RESULT ANALYSIS

A. Dataset Selection

In network traffic classification, datasets and evaluation criteria are not consistent. Many researchers use private self-generated datasets for verification. Although self-generated datasets are relatively easy to obtain label information, their own use of self-generated datasets causes the problem that different algorithms cannot be compared. Therefore, in this paper, we choose the ISCX VPN dataset published by the University of New Brunswick [29].

This dataset divides network packets captured at the data link layer into different pcap files based on the application they were generated using (e.g., Gmail, Skype, Facebook, etc.) and the different activities performed by the application during network communication (e.g., chat, file transfer, or video call, etc.).

B. Evaluation Metrics

We used four evaluation metrics: precision(P_c), recall(R_c), F1 score($F1_c$), and accuracy(acc). Since encrypted traffic classification is a multi-category task, we need to calculate the above metrics for each category separately. Specially, we use N as the total number of training samples; TP_c to indicate the quantity that originally belongs to category c , and is predicted by the model as c ; FP_c indicates the quantity that originally does not belong to category c , but is predicted by the model as c ; TN_c indicates the quantity that does not belong to category c , and is not predicted to be

TABLE II
COMPARISON WITH STATE-OF-THE-ART METHODS.

Method	Accuracy	Precision	Recall	F1-Score
FlowPrint[18]	0.8767	0.6697	0.6651	0.6531
AppScanner[19]	0.6266	0.4864	0.5198	0.4935
BIND[20]	0.6767	0.5152	0.5153	0.4965
DF[r32]	0.6116	0.5706	0.4752	0.4799
FS-Net[9]	0.6647	0.4819	0.4848	0.4737
Tree-RNN[26]	0.9898	0.9898	0.9897	0.9852
Deep Packet[8]	0.9758	0.9785	0.9745	0.9765
PERT[12]	0.8229	0.7092	0.7173	0.6992
CBD[28]	0.8388	0.8849	0.8381	0.8397
ET-BERT(flow)[13]	0.8519	0.7508	0.7294	0.7306
ET-BERT(packet)[13]	0.9962	0.9936	0.9947	0.9941
BFCN[14]	0.9965	0.9936	0.9947	0.9941
Bi-ETC	0.9970	0.9934	0.9951	0.9943

class c ; FN_c indicates the quantity that it belongs to class FN_c , but is misclassified to other class. Hence, the definition of aforementioned four evaluation metrics can be given by

$$P_c = \frac{TP_c}{TP_c + FP_c} \quad (4)$$

$$R_c = \frac{TP_c}{TP_c + FN_c} \quad (5)$$

$$F1_c = \frac{2P_cR_c}{P_c + R_c} \quad (6)$$

$$acc = \frac{TP_c + TN_c}{TP_c + TN_c + FP_c + FN_c} \quad (7)$$

C. Experimental setup and result analysis

1) Implementation details

This experiment was performed with eight Tesla T4 Gpus. The model parameters are set as follows: We set the number of epochs to 15, the initial learning rate to 2×10^{-5} , and dynamically adjust the learning rate with the training process, the ratio of warmup is set to 0.1, the batch size is set to 32, and the dropout is set to 0.1. BERT input sequence length is set to 128, BERT hidden size and BiLSTM hidden size are set to 768, BiLSTM layer number is set to 2, and cross-entropy loss function is used for loss calculation. AdamW was used as the optimizer.

2) Comparison with State-of-the-art Methods

We compare the proposed Bi-ETC model with various state-of-the-art methods, including (1) fingerprint-based construction method: FlowPrint; (2) machine learning-based methods: AppScanner, BIND; (3) Deep learning-based methods: DF, FS-Net, Tree-RNN, Deep Packet; (4) Based on pre-training methods: PERT, CBD, ET-BERT, BFCN.

According to the experimental results in Table 2, the proposed Bi-ETC model can extract and learn discriminative fea-

tures well, can successfully distinguish each application, and outperforms other models on the ISCX VPN dataset. Moreover, the overall performance of the proposed model is higher improved compared to previous models due to the following two reasons.

- After BERT extracts the general features of the encrypted traffic, use the hidden layer state of BiLSTM to record the relevant feature information of the front and rear Token sequences, so as to learn the long-distance dependencies in the Token sequences.
- By simultaneously inserting packet-level Token semantic features at the forward and backward starting positions of BiLSTM, this helps the BiLSTM model capture packet-level features and contextual relationships.

3) Controlled Experiment

1. BERT-BiLSTM VS. BERT-FFNN VS. BERT-LSTM

In our experiments, we use two control models:

(1) BERT-FFNN: This model is the pre-trained model for our experiments [5] and employs BERT-Feed forward neural networks (FFNN), which can provide a benchmark to evaluate the performance of our model in capturing the contextual sequence features of encrypted traffic tokens.

(2) BERT-LSTM: In order to verify the superiority of BiLSTM forward learning and backward learning, we choose BERT-LSTM as another control experimental model.

Since BERT-FFNN is the model selected in our pre-training stage, and ISCX VPN dataset is also used in the pre-trained large-scale unlabeled dataset. Therefore, at the beginning, the model parameters of BERT-FFNN have fallen within a good performance range. However, after the ninth epoch, we observe that the model has reached the convergence point, the Accuracy basically does not change, and the improvement of the model performance by further training is no longer significant. On the contrary, continued training may lead to

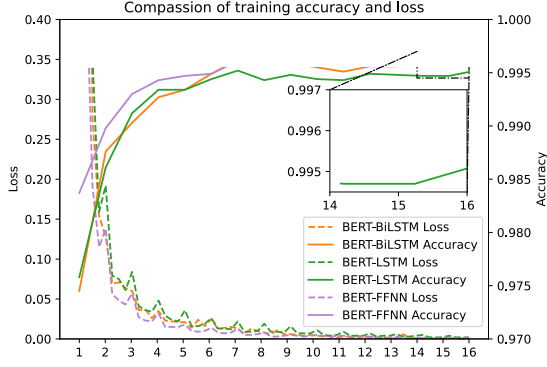


Fig. 2. Comparison of accuracy and loss values in control experiments.

overfitting. This observation suggests that further tuning the model or exploring other optimization strategies may be a more promising avenue to improve performance.

Since BERT-FFNN only inputs [CLS]Token features extracted by BERT into FFNN for classification, there are a lot of Token features left unclassified, while BERT-biLSTM can capture complex patterns between tokens. To prevent overfitting, we use dropout technique, so the accuracy fluctuates during the period, but finally surpasses BERT-FFNN. BERT-LSTM, on the other hand, forgets the most important [CLS] packet-level features and thus performs even worse than FFNN.

4) Ablation Experiment

We conducted two ablation experiments on Bi-LSTM to verify the feasibility and advantages of the Bi-ETC.

1. [CLS]-tokens-[CLS] VS. [CLS]-tokens;

(1) [CLS]-tokens-[CLS] : This is the approach described in Bi-ETC, where the sequence of [CLS]tokens in the head is copied to the end of the BiLSTM input sequence.

(2) [CLS]-tokens: It is all sequence features extracted by BERT, including [CLS]token and all encrypted traffic token sequences after it, and input them all into BiLSTM.

We observe that our Loss decreases faster and we end up

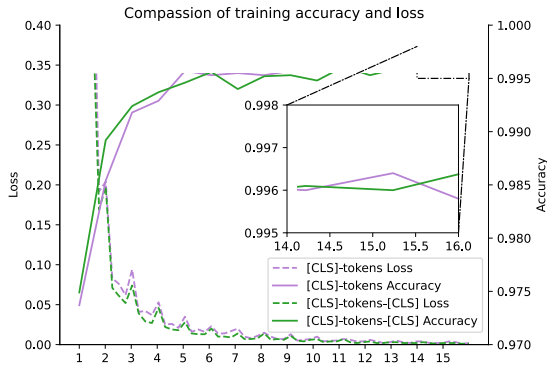


Fig. 3. Comparison of accuracy and loss values between [CLS]-token and [CLS]-token-[CLS].

with the highest accuracy. This is because, for the lookahead LSTM sequence, the sequence captures the most important [CLS]Token feature sequence very early; For the backward LSTM sequence, if we do not copy the [CLS]Token sequence of the head to the end of the entire feature sequence, then the backward LSTM sequence needs to be at the end to extract the [CLS] feature sequence. So we consider inserting a [CLS]Token feature sequence before and after the input vector of the Bi-LSTM model. In other words, the output vector of the one-before and one-after corresponding [CLS] in the last hidden layer is used as the semantic feature representation of all traffic, which helps the BiLSTM model to capture the packetlevel features and context, and provides a more comprehensive traffic classification representation.

2.tanh VS. PReLU

In this experiment, we selected two activation functions for comparison. The first time we use tanh as the activation function, the curves of tanh and sigmoid function are relatively similar, the difference is the output interval, the output interval of tanh is between (-1,1), and the whole function is centered at 0. However, using tanh function as the activation function in the BiLSTM model itself has the problem of gradient disappearance. PReLU scales each negative element value by a factor, which not only plays the role of nonlinear transformation, but also uses a different scaling factor for each channel on the negative axis, which plays a role similar to the attention mechanism. The output vector of Bi-LSTM is fed into a classifier with PReLU as the activation function to obtain the final prediction result.

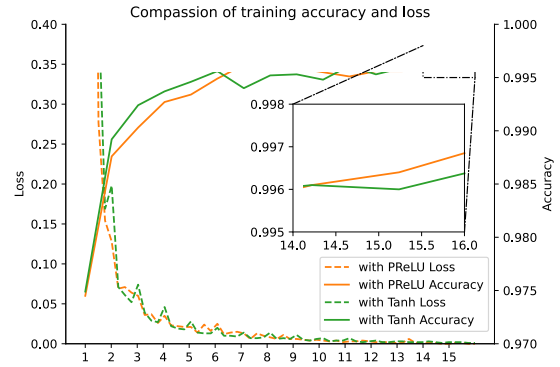


Fig. 4. Comparison of accuracy and loss values using different activation functions tanh and PReLU.

5) Performance of different application recognition tasks

Table 3 shows the performance achieved by different models on the application recognition task on the test set. We chose two effect pre-trained models ET-BERT [5] and BFCN [6] to compare with Bi-ETC on application task recognition. We have achieved the best results in Precision, Recall, and F1 on most of the classification samples, and we have also achieved good recognition results on the 823 ICQ small samples.

Bi-ETC is able to make more comprehensive use of the information in the packet in the encrypted traffic classification

TABLE III
PERFORMANCE COMPARISON OF BI-ETC WITH ET-BERT AND BFCN IN THE APPLICATION IDENTIFICATION TASK.

Class	Ours			ET-BERT			BFCN		
	Precision	Recall	F1-Score	Precision	Recall	F1-Score	Precision	Recall	F1-Score
AIM_Chat	0.978	0.978	0.978	0.997	0.970	0.974	0.992	0.978	0.985
Email	0.994	1.000	0.997	0.994	1.000	0.997	0.994	1.000	0.997
Facebook	0.998	0.992	0.995	0.996	0.984	0.990	0.992	0.990	0.991
Gmail	0.992	0.998	0.995	0.986	0.994	0.990	0.986	0.994	0.990
Hangout	0.996	0.996	0.996	0.996	0.998	0.997	1.000	0.996	0.998
ICQ	0.941	0.976	0.958	0.920	0.976	0.947	0.941	0.976	0.958
Netflix	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
SCP	1.000	0.994	0.997	1.000	0.994	0.997	1.000	0.994	0.997
Skype	0.996	0.996	0.996	0.998	0.998	0.998	0.994	0.998	0.996
Spotify	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Tor	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Torrent	0.994	1.000	0.997	0.992	1.000	0.996	0.992	0.998	0.995
Vimeo	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
VoipBuster	1.000	0.990	0.995	1.000	0.998	0.994	1.000	0.998	0.994
VPN-FTPS	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
VPN-SFTP	1.000	0.998	0.999	1.000	0.998	0.999	1.000	0.998	0.999
Youtube	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

task, and is able to consider the context of each tag, helping us to better understand and distinguish between different types of applications.

In addition, the confusion matrix is usually applied to validate the results of the classification, to observe the performance of Bi-ETC on various categories, and to provide a better diagram to understand how well Bi-ETC is working, the performance confusion matrix of Bi-ETC is shown in Figure 8. The dark elements on the main diagonal show that the Bi-ETC model performs well on the classification of each application with little confusion error.

VI. CONCLUSION

In this paper, we introduce suitable BERT and BiLSTM models, and propose a novel encrypted traffic feature extraction and traffic classification method to solve the problem that pre-trained models are difficult to capture long-distance dependencies between traffic sequence features. In Bi-ETC, we use BERT to generate feature representations of encrypted traffic tokens, and capture the contextual semantic features between tokens by BiLSTM. Bi-ETC achieves satisfactory performance on the widely used ISCX-VPN dataset, with an F1 score of 99.43% and an accuracy of 99.7% respectively. In the future, we will further expand and deeply study the scope of encrypted traffic data to mine more traffic feature representations. We plan to apply our method to more traffic classification tasks to evaluate its generality and effect on different scenarios and datasets. In addition, we plan to introduce an attention mechanism on the output of BiLSTM to assist and enhance the classification effect. Such an attention mechanism can make the model pay more attention to important tokens and

features, thereby improving the performance and robustness of the model.

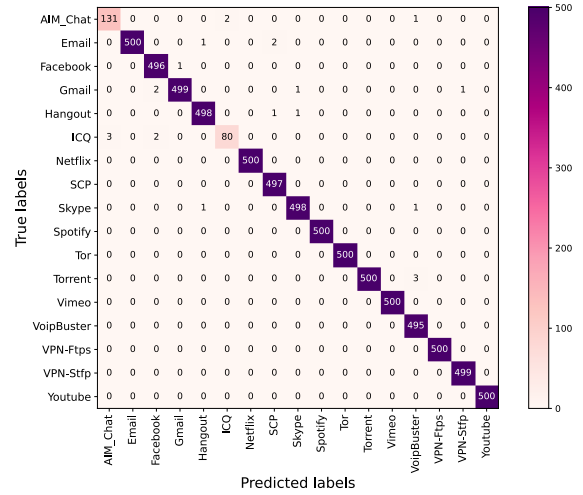


Fig. 5. Confusion matrix for TETBB model performance.

ACKNOWLEDGMENT

This work was supported in The Major Key Project of PCL (Grant No. PCL2022A03), the National Key Research and Development Program of China (2021YFB2012402), The National Natural Science Foundation of China (Grant No. 61976064). Corresponding author: Ning Hu (e-mail: hun@pcl.ac.cn)

REFERENCES

- [1] Keke Gai, Meikang Qiu, and Hui Zhao. "Privacy-preserving data encryption strategy for big data in mobile cloud computing". In: *IEEE Transactions on Big Data* 7.4 (2017), pp. 678–688.
- [2] Liangchen Chen et al. "Research status and development trends on network encrypted traffic identification". In: *Netinfo Secur.(03)* (2019), pp. 19–25.
- [3] Fannia Pacheco et al. "Towards the deployment of machine learning solutions in network traffic classification: A systematic survey". In: *IEEE Communications Surveys & Tutorials* 21.2 (2018), pp. 1988–2014.
- [4] Alberto Dainotti, Antonio Pescapé, and Kimberly C Claffy. "Issues and future directions in traffic classification". In: *IEEE network* 26.1 (2012), pp. 35–40.
- [5] Soheil Hassas Yeganeh et al. "Cute: Traffic classification using terms". In: *2012 21st International Conference on Computer Communications and Networks (ICCCN)*. IEEE. 2012, pp. 1–9.
- [6] Vincent F Taylor et al. "Robust smartphone app identification via encrypted network traffic analysis". In: *IEEE Transactions on Information Forensics and Security* 13.1 (2017), pp. 63–78.
- [7] Khaled Al-Naami et al. "Adaptive encrypted traffic fingerprinting with bi-directional dependence". In: *Proceedings of the 32nd Annual Conference on Computer Security Applications*. 2016, pp. 177–188.
- [8] Mohammad Lotfollahi et al. "Deep packet: A novel approach for encrypted traffic classification using deep learning". In: *Soft Computing* 24.3 (2020), pp. 1999–2012.
- [9] Chang Liu et al. "Fs-net: A flow sequence network for encrypted traffic classification". In: *IEEE INFOCOM 2019-IEEE Conference On Computer Communications*. IEEE. 2019, pp. 1171–1179.
- [10] Jacob Devlin et al. "Bert: Pre-training of deep bidirectional transformers for language understanding". In: *arXiv preprint arXiv:1810.04805* (2018).
- [11] Alexey Dosovitskiy et al. "An image is worth 16x16 words: Transformers for image recognition at scale". In: *arXiv preprint arXiv:2010.11929* (2020).
- [12] Hong Ye He, Zhi Guo Yang, and Xiang Ning Chen. "PERT: Payload encoding representation from transformer for encrypted traffic classification". In: *2020 ITU Kaleidoscope: Industry-Driven Digital Transformation (ITU K)*. IEEE. 2020, pp. 1–8.
- [13] Xinjie Lin et al. "Et-bert: A contextualized datagram representation with pre-training transformers for encrypted traffic classification". In: *Proceedings of the ACM Web Conference 2022*. 2022, pp. 633–642.
- [14] Zhaolei Shi et al. "BFCN: A Novel Classification Method of Encrypted Traffic Based on BERT and CNN". In: *Electronics* 12.3 (2023), p. 516.
- [15] Petr Velan et al. "A survey of methods for encrypted traffic classification and analysis". In: *International Journal of Network Management* 25.5 (2015), pp. 355–374.
- [16] Alberto Dainotti, Antonio Pescapé, and Kimberly C Claffy. "Issues and future directions in traffic classification". In: *IEEE network* 26.1 (2012), pp. 35–40.
- [17] Justine Sherry et al. "Blindbox: Deep packet inspection over encrypted traffic". In: *Proceedings of the 2015 ACM conference on special interest group on data communication*. 2015, pp. 213–226.
- [18] Thijs Van Ede et al. "Flowprint: Semi-supervised mobile-app fingerprinting on encrypted network traffic". In: *Network and Distributed System Security Symposium (NDSS)*. Vol. 27. 2020.
- [19] Vincent F Taylor et al. "Robust smartphone app identification via encrypted network traffic analysis". In: *IEEE Transactions on Information Forensics and Security* 13.1 (2017), pp. 63–78.
- [20] Khaled Al-Naami et al. "Adaptive encrypted traffic fingerprinting with bi-directional dependence". In: *Proceedings of the 32nd Annual Conference on Computer Security Applications*. 2016, pp. 177–188.
- [21] Giuseppe Aceto et al. "Know your big data trade-offs when classifying encrypted mobile traffic with deep learning". In: *2019 Network traffic measurement and analysis conference (TMA)*. IEEE. 2019, pp. 121–128.
- [22] Shahbaz Rezaei and Xin Liu. "Deep learning for encrypted traffic classification: An overview". In: *IEEE communications magazine* 57.5 (2019), pp. 76–81.
- [23] Zhanyi Wang. "The applications of deep learning on traffic identification". In: *BlackHat USA* 24.11 (2015), pp. 1–10.
- [24] Yi Zeng et al. "Deep – Full – Range: a deep learning based network encrypted traffic classification and intrusion detection framework". In: *IEEE Access* 7 (2019), pp. 45182–45190.
- [25] Haipeng Yao et al. "Identification of encrypted traffic through attention mechanism based long short term memory". In: *IEEE transactions on big data* 8.1 (2019), pp. 241–252.
- [26] Xinming Ren, Huaxi Gu, and Wenting Wei. "Tree-RNN: Tree structural recurrent neural network for network traffic classification". In: *Expert Systems with Applications* 167 (2021), p. 114363.
- [27] Redvan Ghasemlounia et al. "Developing a novel framework for forecasting groundwater level fluctuations using Bi-directional Long Short-Term Memory (BiLSTM) deep neural network". In: *Computers and Electronics in Agriculture* 191 (2021), p. 106568.
- [28] Xinyi Hu et al. "CBD: A deep-learning-based scheme for encrypted traffic classification with a general pre-training method". In: *Sensors* 21.24 (2021), p. 8231.
- [29] Payap Sirinam et al. "Deep fingerprinting: Undermining website fingerprinting defenses with deep learning". In: *Proceedings of the 2018 ACM SIGSAC Conference on Computer and Communications Security*. 2018, pp. 1928–1943.